4.4.4 Cumulative frequency curve (Ogive)

Cumulative frequency curve (Ogive) is drawn to represent the cumulative frequency distribution. There are two types of Ogives such as 'less thanOgive curve' and 'more thanOgive curve'. To draw these curves, we have to calculate the 'less than' cumulative frequencies and 'more than' cumulative frequencies. The following procedure can be followed to draw the ogive curves:

Less than Ogive: Less than cumulative frequency of each class is marked against the corresponding upper limit of the respective class. All the points are joined by a free-hand curve to draw **the less than ogive** curve.

More than Ogive: More than cumulative frequency of each class is marked against the corresponding lower limit of the respective class. All the points are joined by a free-hand curve to draw the **more than ogive** curve.

Both the curves can be drawn separately or in the same graph. If both the curves are drawn in the same graph, then the value of abscissa (x-coordinate) in the point of intersection is the median.

Median is a measure of central tendency, which divides the given data/distribution into two equal parts. It is discussed much in detail in Unit V

If the curves are drawn separately, median can be calculated as follows:

Draw a line perpendicular to Y-axis at $y=N/2$. Let it meet the Ogive at C. Then, draw a perpendicular line to X-axis from the point C. Let it meet the X-axis at M. The abscissa of M is the median of the data.

Example 4.14

Draw the less than Ogive curve for the following data:

Daily Wages (in Rs.)	70- 80	80- 90	90-100	100-110	110-120	120-130	130-140	140-150
No. of workers	12	18	35	42	50	45	20	8

Also, find

(i) The Median

(ii) The number of workers whose daily wages are less than ₹ 125.

Solution:

Since we are displaying the distribution of Daily Wages and No. of workers, the Ogive curve is drawn, to provide better understanding about the wages and No. of workers.